

AKASH R M

AI Developer | Computer Vision & Deep Learning Engineer
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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

AI Developer with hands-on experience building real-time computer vision systems and production-grade, full-stack AI applications. Specialized in deep learning, face recognition, and multi-object tracking, with a proven record of shipping end-to-end pipelines – from model optimization and low-latency video inference to REST/WebSocket APIs and deployed web and mobile frontends. Strong in Python, FastAPI, React, and GPU-accelerated inference using ONNX Runtime and CUDA.

TECHNICAL SKILLS

- **Languages:** | Python | TypeScript | JavaScript (ES2022) | SQL | HTML5 | CSS3 |
- **AI / Computer Vision:** | Deep Learning | Machine Learning | Computer Vision | Face Detection & Recognition | Object Detection | Multi-Object Tracking (MOT) | Action Recognition | Image Classification | Model Optimization |
- **ML Frameworks:** | PyTorch | TensorFlow | Keras | InsightFace (SCRFD, ArcFace) | ONNX Runtime (GPU / CUDA) | OpenCV | NumPy | Pillow | YOLO |
- **Backend:** | FastAPI | Flask | Uvicorn (ASGI) | Pydantic | SQLAlchemy (ORM) | RESTful API Design | WebSocket APIs | JWT Auth | bcrypt |
- **Frontend:** | React 19 | TanStack Router | TanStack Query | React Hook Form | Zod | Tailwind CSS | Radix UI | Recharts | Vite |
- **Video & Streaming:** | RTSP | WebRTC (aiortc) | PyAV / FFmpeg | MJPEG | MSE / fMP4 | ONVIF | NVDEC / libx264 | Live Multi-Camera Pipelines |
- **Databases & Cloud:** | PostgreSQL | SQLite | Supabase | AWS (EC2) | Railway | Cloudflare |
- **DevOps & Tools:** | Docker | Git | GitHub | CI/CD | Capacitor (Android) | Linux |

EXPERIENCE

AI Developer – GROW AI (iQue Ventures Pvt. Ltd.), Bengaluru 2026 – Present

- Built real-time person detection and multi-object tracking systems for video analytics, processing live RTSP camera streams with optimized GPU inference.
- Engineered a production facial-recognition attendance platform, end to end – face detection/recognition pipeline, FastAPI backend, PostgreSQL data layer, and React + TypeScript dashboard.
- Improved detection robustness and recognition accuracy through threshold tuning, embedding-quality validation, and tracking refinement.
- Deployed and maintained low-latency live video streaming via WebRTC and MSE/fMP4 across multiple concurrent cameras.

AI Developer – Owlytics (iQue Ventures Pvt. Ltd.), Bengaluru 2025 – 2026

- Designed and implemented deep learning models for detection and recognition tasks.
- Built scalable ML pipelines spanning preprocessing, training, evaluation, and deployment.
- Integrated trained AI models into production backend systems and APIs.

Assistant Technical Writer – Rpinnacle Research Solutions, Tamil Nadu 2024 – 2025

- Produced technical documentation for deep learning and data science research projects.
- Contributed to dataset analysis, experimental evaluation, and comparative model studies.

PROJECTS

Real-Time Facial Recognition Attendance Platform 2026

- Full-stack workforce platform: automated attendance from live camera face recognition, payroll computation, and HR/admin dashboards.
- Backend in FastAPI + SQLAlchemy + PostgreSQL; GPU face recognition with InsightFace + ONNX Runtime

CUDA; JWT/bcrypt role-based auth.

- Live multi-camera streaming via WebRTC, aiortc, and PyAV, plus a React 19 + TanStack + Tailwind frontend and a Capacitor Android app.
- Containerized with Docker and deployed on Railway with Cloudflare tunneling.

AI Real-Time Person Detection & Tracking System

2026

- Built a real-time person detection and multi-object tracking pipeline using YOLO, processing RTSP streams with optimized inference.
- Implemented frame-wise analytics and unique person counting.

AI Cricket Shot Classification & Batting Analysis System

2026

- Developed a cricket shot classifier using a CNN-LSTM architecture with MobileNetV2 for video-based action recognition.
- Designed preprocessing/feature-engineering pipelines and backend APIs for confidence scoring and performance tiering.

Vision Snap – Computer Vision Dataset Tool

2025

- Built a dataset-generation tool in React with webcam capture, video frame extraction, and automated dataset packaging workflows.

ALL IN ONE – OCR Web Application

2025

- Built a JavaScript OCR web app with a modular architecture and responsive UI for structured, multi-page text and image extraction.

Hospital Management System with Brain Tumor Detection

2024

- Django-based hospital management system with authentication workflows and integrated deep-learning tumor detection and medical reporting.

INTERNSHIPS & CERTIFICATIONS

- **Python Programming Intern** – Srishti Innovations, Technopark, Kerala

July 2023

- **In-Plant Training** – iTrobes Technologies Pvt. Ltd., Tamil Nadu

July 2022

EDUCATION

B.E. Computer Science and Engineering

Bethlahem Institute of Engineering, Tamil Nadu

2020 – 2024